

BLINK-IT SQL ANALYTICS PROJECT REPORT

1. Introduction

The Blinkit SQL Analytics Project focuses on analyzing grocery delivery and retail operations using SQL and relational database concepts. The project uses multiple datasets related to customers, orders, products, and delivery performance to generate meaningful business insights.

The objective of this project is to perform real-world business analysis using SQL queries and identify patterns in customer behavior, product sales, revenue generation, and delivery efficiency.

2. Objectives

- Analyze customer purchasing behavior
- Identify high-value customers
- Analyze product and category sales performance
- Evaluate delivery efficiency and operational performance
- Study customer segmentation and spending trends
- Generate business insights using SQL queries

3. Tools & Technologies Used

- MySQL Workbench
- SQL
- CSV/XLS Datasets
- Relational Database Concept

4. Database Design

The project database was created using MySQL and consists of the following tables:

Tables Used

- customers
- orders
- order_items
- products
- delivery_performance

Relationship Flow

customers → orders → order_items → products

orders → delivery_performance

SQL Concepts Used

- JOIN Operations
- Aggregate Functions
- GROUP BY
- HAVING Clause
- ORDER BY
- Window Functions
- Subqueries
- Business KPI Analysis

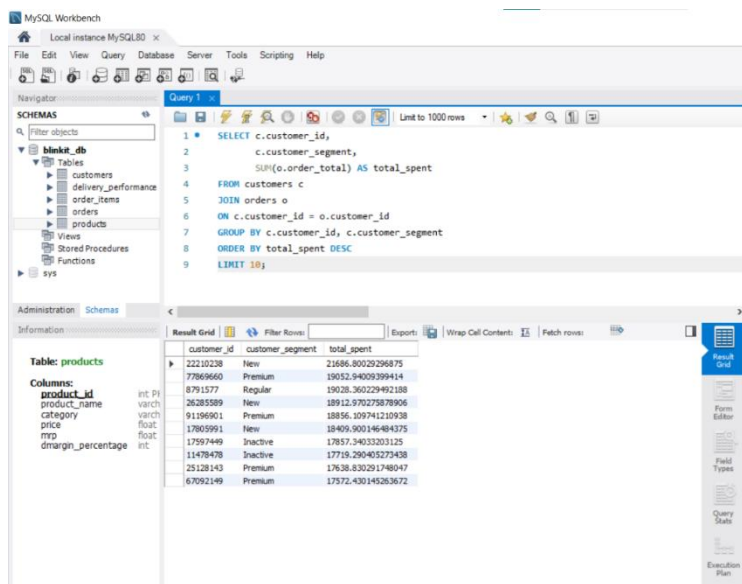
6. Business Analysis

❖ Top High-Value Customers

Objective: To identify customers contributing the highest revenue to the business

SQL Query:

```
SELECT c.customer_id,  
       c.customer_segment,  
       ROUND(SUM(o.order_total), 2) AS total_spent  
FROM customers c  
JOIN orders o  
ON c.customer_id = o.customer_id  
GROUP BY c.customer_id, c.customer_segment  
ORDER BY total_spent DESC  
LIMIT 10;
```



The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying the SQL query. The 'Result Grid' shows the results of the query, which are 10 rows of customer data. The columns are customer_id, customer_segment, and total_spent. The results are sorted by total_spent in descending order.

customer_id	customer_segment	total_spent
22210238	New	21686.80029296875
77869660	Premium	19052.94009399414
8791577	Regular	19028.360229492188
26285589	New	18912.970275878906
91196901	Premium	15856.109741210938
17805991	New	18409.900146484375
17597449	Inactive	17857.34032031125
11478478	Inactive	17719.290405277438
25128143	Premium	17638.830291748047
67092149	Premium	17572.430149263672

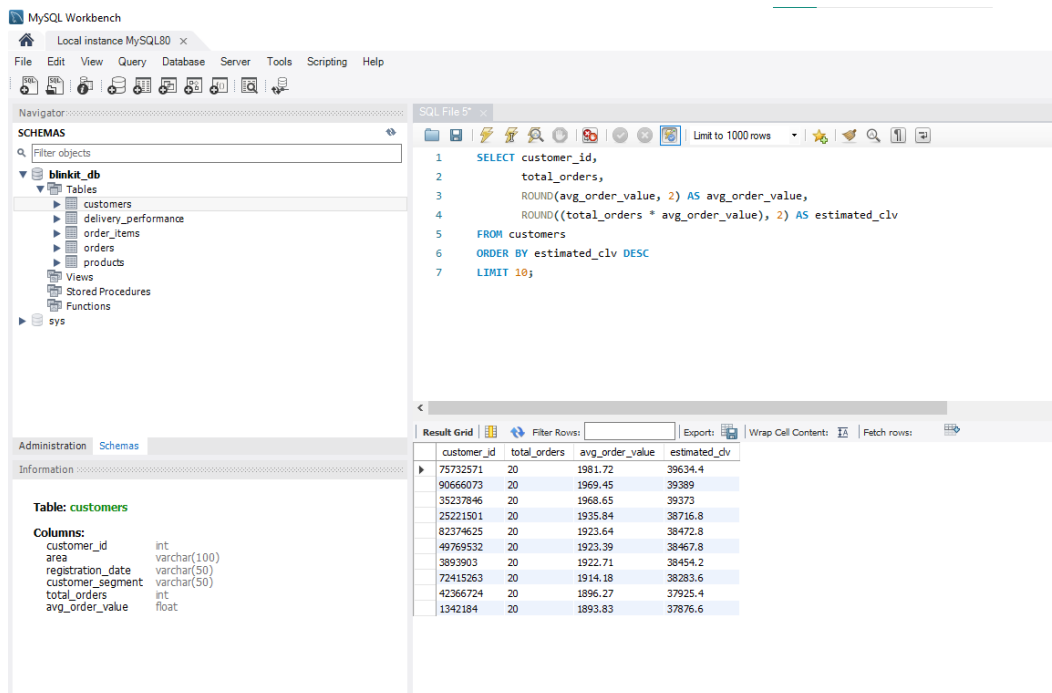
Business Insight: Premium and high-engagement customers contribute significantly to total business revenue. These customers can be targeted through loyalty programs and personalized offers.

❖ Customer Lifetime Value Estimation

Objective: To estimate the long-term value of customers based on spending behavior and order frequency.

SQL Query:

```
SELECT customer_id,  
       total_orders,  
       ROUND(avg_order_value, 2) AS avg_order_value,  
       ROUND((total_orders * avg_order_value), 2) AS estimated_clv  
FROM customers  
ORDER BY estimated_clv DESC  
LIMIT 10;
```



The screenshot displays the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'blinkit_db' database with various tables and views. The central SQL editor contains the query to calculate the estimated customer lifetime value (CLV) for the top 10 customers. The bottom pane shows the 'Result Grid' with the following data:

customer_id	total_orders	avg_order_value	estimated_clv
75732571	20	1981.72	39634.4
90666073	20	1969.45	39389
35237846	20	1968.65	39373
25221501	20	1935.84	38716.8
82374625	20	1923.64	38472.8
49769532	20	1923.39	38467.8
3893903	20	1922.71	38454.2
72415263	20	1914.18	38283.6
42366724	20	1896.27	37925.4
1342184	20	1893.83	37876.6

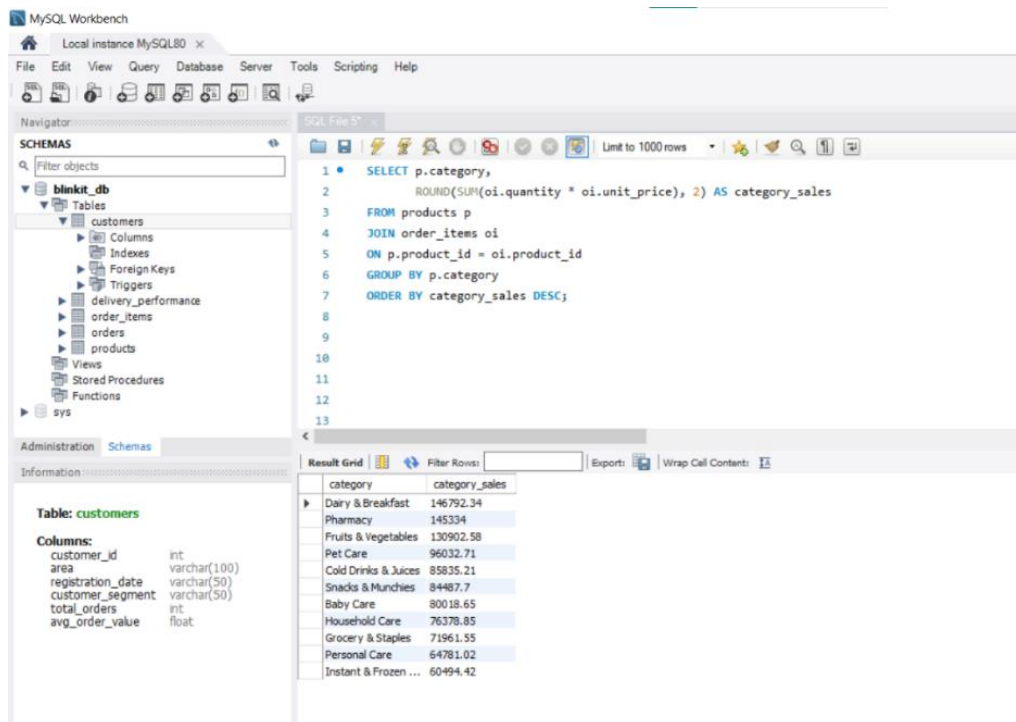
Business Insight: Customers with higher estimated lifetime value are strategically important for long-term profitability and customer retention initiatives.

❖ Category-wise Sales Performance

Objective: To evaluate sales performance across different product categories.

SQL Query:

```
SELECT p.category,  
       ROUND(SUM(oi.quantity * oi.unit_price), 2) AS category_sales  
FROM products p  
JOIN order_items oi  
ON p.product_id = oi.product_id  
GROUP BY p.category  
ORDER BY category_sales DESC;
```



The screenshot displays the MySQL Workbench interface. On the left, the 'SCHEMAS' pane shows the 'blinkit_db' database with various tables like 'customers', 'delivery_performance', 'order_items', 'orders', 'products', 'views', 'stored_procedures', and 'functions'. The 'products' table is selected. The main editor shows the SQL query for category-wise sales performance. The 'Result Grid' at the bottom displays the query results, sorted by category_sales in descending order.

category	category_sales
Dairy & Breakfast	146792.34
Pharmacy	145334
Fruits & Vegetables	130902.58
Pet Care	96032.71
Cold Drinks & Juices	85835.21
Snacks & Munchies	84487.7
Baby Care	80018.65
Household Care	76378.85
Grocery & Staples	71961.55
Personal Care	64781.02
Instant & Frozen ...	60494.42

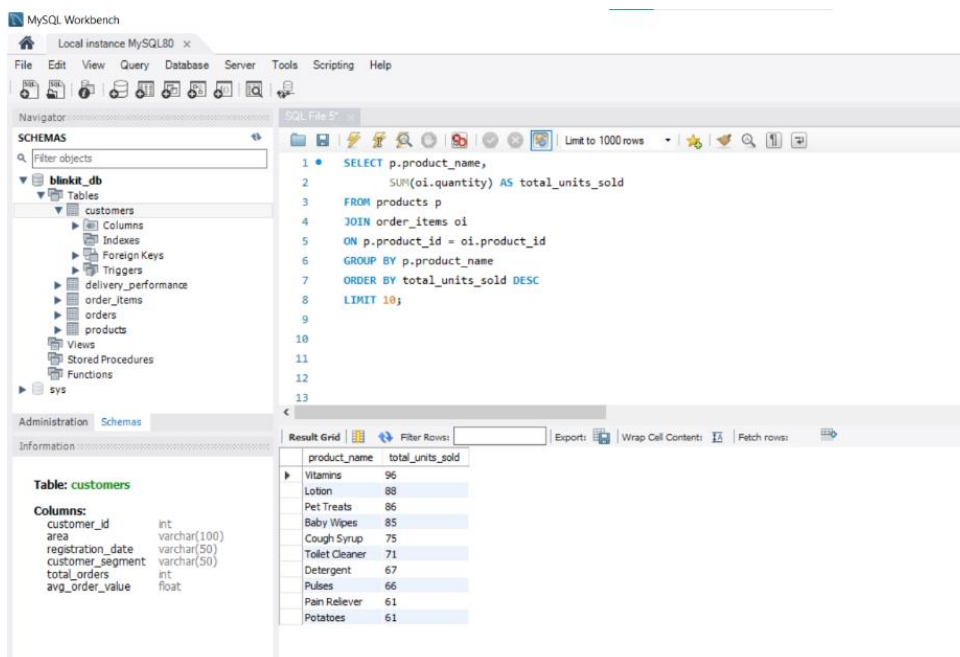
Business Insight: Top-performing categories contribute significantly to overall business revenue and require focused inventory and operational management.

❖ Top Revenue Generating Products

Objective: To identify products generating the highest revenue for the business.

SQL Query:

```
SELECT p.product_name,  
       p.category,  
       ROUND(SUM(oi.quantity * oi.unit_price), 2) AS total_revenue  
FROM order_items oi  
JOIN products p  
ON oi.product_id = p.product_id  
GROUP BY p.product_name, p.category  
ORDER BY total_revenue DESC  
LIMIT 10;
```



The screenshot shows the MySQL Workbench interface. The left sidebar displays the 'SCHEMAS' panel with a tree view of the database structure, including tables like 'customers', 'order_items', and 'products'. The main window shows the SQL query editor with the following query:

```
1 SELECT p.product_name,  
2     SUM(oi.quantity) AS total_units_sold  
3 FROM products p  
4 JOIN order_items oi  
5 ON p.product_id = oi.product_id  
6 GROUP BY p.product_name  
7 ORDER BY total_units_sold DESC  
8 LIMIT 10;
```

Below the query editor, the 'Result Grid' tab is active, displaying the results of the query in a table format:

product_name	total_units_sold
Vitamins	96
Lotion	88
Pet Treats	86
Baby Wipes	85
Cough Syrup	75
Toilet Cleaner	71
Detergent	67
Pulses	66
Pain Reliever	61
Potatoes	61

Business Insight: High-performing products should receive priority in inventory planning, promotions, and marketing campaigns.

❖ Orders Above Average Order Value

Objective: To identify orders exceeding the average business order value.

SQL Query:

```
SELECT order_id,  
  
       customer_id,  
  
       order_total  
  
FROM orders  
  
WHERE order_total >  
  
(  
  
   SELECT AVG(order_total)  
  
   FROM orders  
  
);
```

The screenshot shows the MySQL Workbench interface. The SQL Editor contains the query to find orders above the average order total. The Results window displays a list of orders with their IDs, customer IDs, and totals. The left sidebar shows the database schema for 'blinkit_db'.

Table: orders

Columns:

- order_id: varchar(50)
- customer_id: int
- promised_delivery_time: varchar(50)
- actual_delivery_time: varchar(50)
- delivery_status: varchar(50)
- order_total: float
- payment_method: varchar(50)
- delivery_partner_id: bigint

order_id	customer_id	order_total
1961864118	30065862	3197.07
5427684290	83208567	2526.68
3265154092	43367112	3161.43
4455336265	43431724	2362.33
5504941865	76512976	2287.68
2137046916	12059049	3373.96
2899700050	84948018	2706.54
6625178717	62068277	5718.09
7169557514	84824280	2970.9
2268306684	66742602	2991.57
1769403463	3292927	2541.32
5003387344	26592009	2475.69
964822979	88520326	5804.23
9347571484	29562072	3138.06
3286177534	10225164	3612.26
6355378227	67092149	5019.02
277376321	89784343	3694.48
5838302640	96795047	3377.44
1380971977	16561710	3117.74
1221099013	35156679	4335.11
5562922993	62416456	2292.64

Business Insight: Orders above average value indicates premium customer purchases and contribute disproportionately to overall revenue growth.

7. Key Insights Generated

- High-value customers contribute a major portion of total revenue
- Certain product categories dominate sales performance
- Customer segments show different spending behaviors
- Digital payment methods are highly preferred by customers
- Delivery efficiency directly impacts operational performance
- Revenue concentration exists among a limited number of products and customers

8. Conclusion

The Blinkit SQL Analytics Project successfully demonstrates the use of SQL for real-world business analysis and decision-making. By analyzing customer, product, sales, and delivery datasets, meaningful insights were generated regarding business performance and operational efficiency.

The project helped strengthen practical understanding of SQL concepts such as joins, aggregations, grouping, window functions, and business KPI analysis. It also demonstrated how SQL can be effectively used for data-driven business analytics in retail and delivery-based industries.